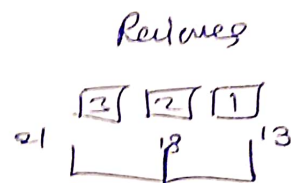
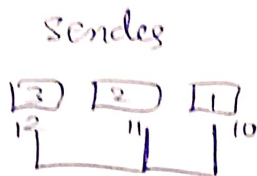


Quality of Service :-

If we want to provide quality for applications running on internet, by considering 4 criteria.

- i] Delay : packet takes additional time to travel in transmission media
- ii] Reliability : Successfully to send the packet to receiver.
- iii] Jitter : uneven delay in packet
- iv] Bandwidth : capacity of channel to carry data towards receiver.



variation in delay in packets at destination while receiving.

Flow Classes :-

Depending on what type of data it transmit, it categorized into 5 classes called as flow classes.

- i] Constant Bit rate (CBR) : In this class applications can able to transfer audio and video to other end.
- ii] Variable Bit-rate - Non real time (VBR-NRT)
- In this class application can able to transfer email
- iii] Variable-Bit rate - Real Time : It is used for communication of images (VBR-RT)
- iv] Fixed Bit Rate - It is used for transmission of text (FBT)
- v] Undefined Bit Rate (UBT) - It is combination of all data can be transferred

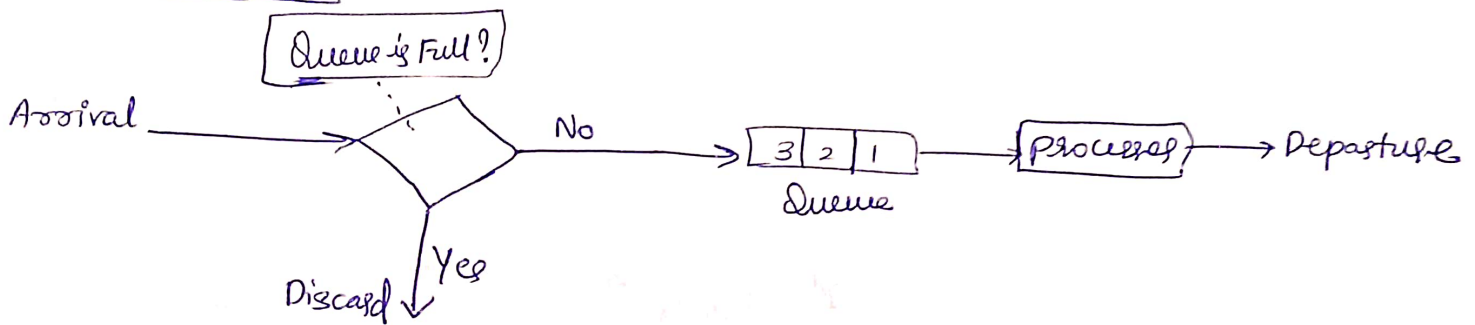
Flow Control :- at application layer to provide quality of service.

~~It~~ It is implemented in Routers

Scheduling Algorithms :-

- i] FIFO scheduling
- ii] Priority scheduling
- iii] Weighted fair scheduling

i] FIFO scheduling :-



→ In FIFO queuing packets wait in buffers until the node is ready to process them. If the average arrival rate is higher than the average processing rate the queue will fill up and new packets will be discarded.

→ FIFO queuing is the default scheduling in the internet

ii] Priority Queuing :-

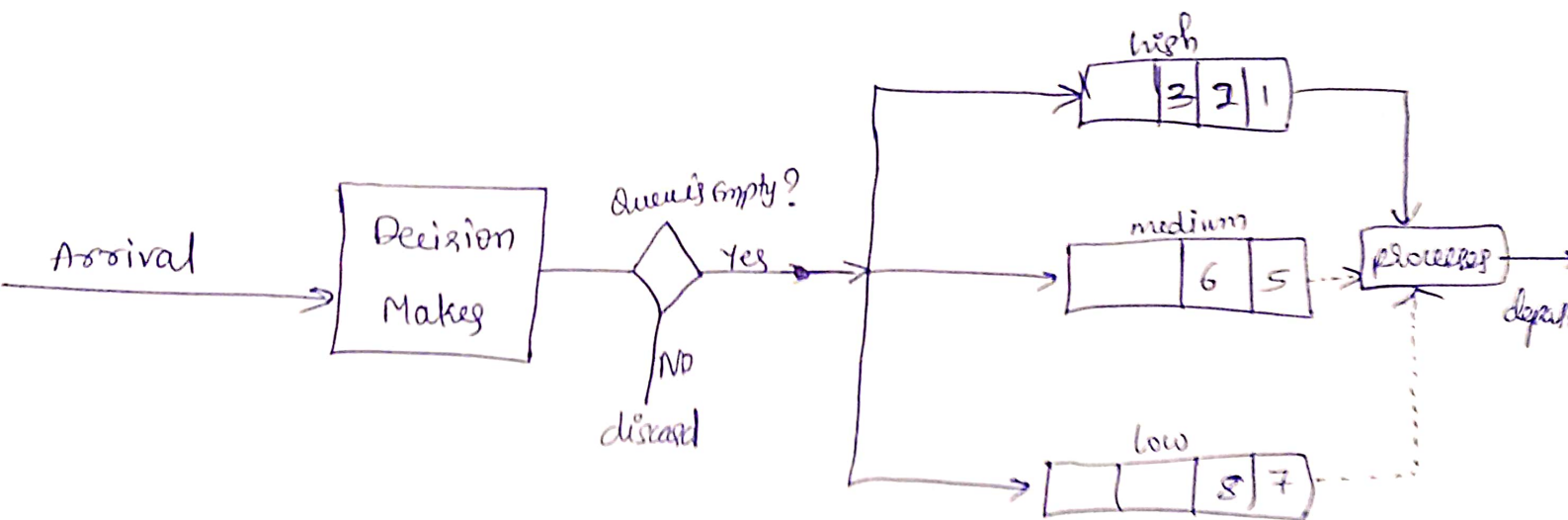
→ Queue delay in FIFO queuing often degrades quality of service in the network. A frame carrying real time packets may have to wait a long time behind a frame carrying a small file.

To overcome this problem priority queuing is introduced.

In priority queuing, packets are first assigned to a priority class. Each priority class has its own queue. The packets in the highest-priority queue are processed first. packets in the lowest priority queue are processed last. The system will not stop serving queue until it is empty.

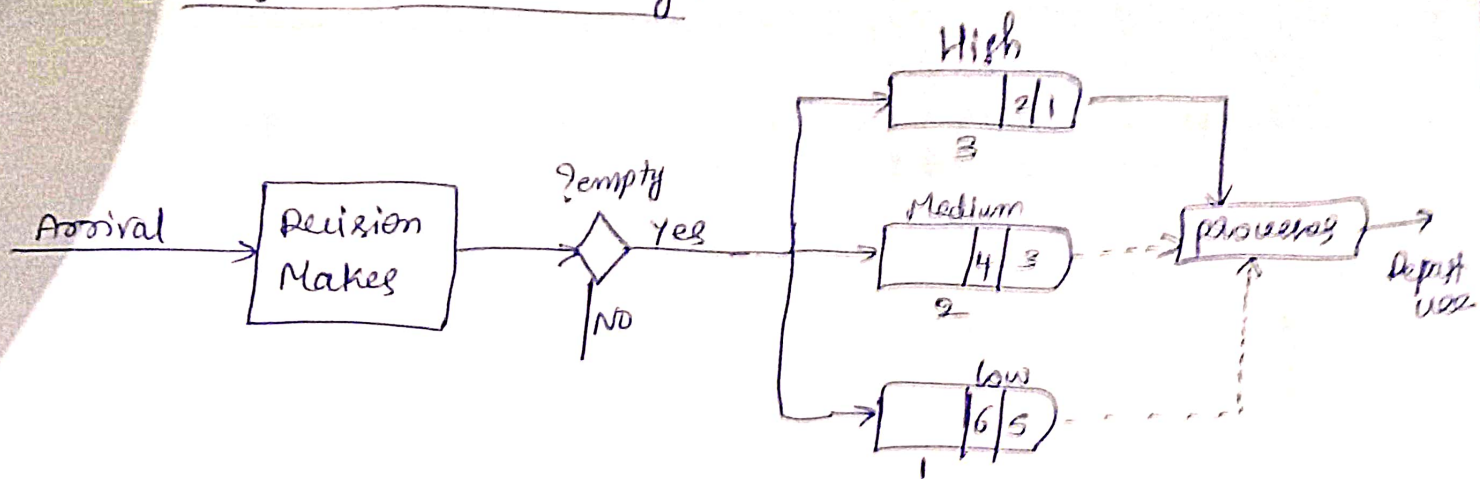
→ A packet priority is determined from a specific field in the packet header.

- The tos field of IPv4 header
- The priority field of IPv6 header.
- A priority number assigned to application (destination port numbers).



disadvantage: packets at low priority queue will be waiting for longer amount of time to processing.

iii) Weighted Fair scheduling:-



In this technique the packets are still assigned to different classes and admitted to different queues. The queues, however, are weighted based on priority of the queues.

High priority $\rightarrow$ higher weight.

The system processes packets in each queue in round robin fashion with number of packets selected from each queue based on corresponding weight.

Traffic Policing as shaping:-

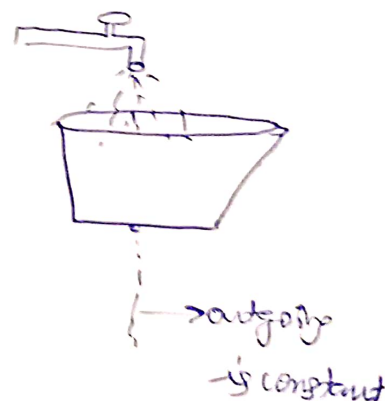
$\downarrow$ rate of outgoing packets $\rightarrow$ rate of incoming packets

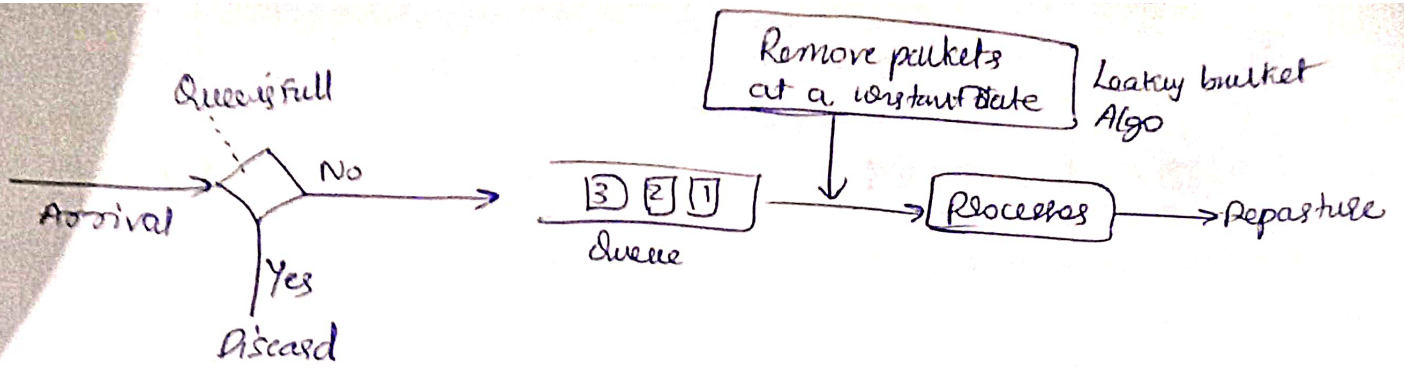
Traffic policing : controlling of outgoing packets at router.

Traffic shaping : controlling of incoming packets at router.

i] Leaky bucket Algorithm:-

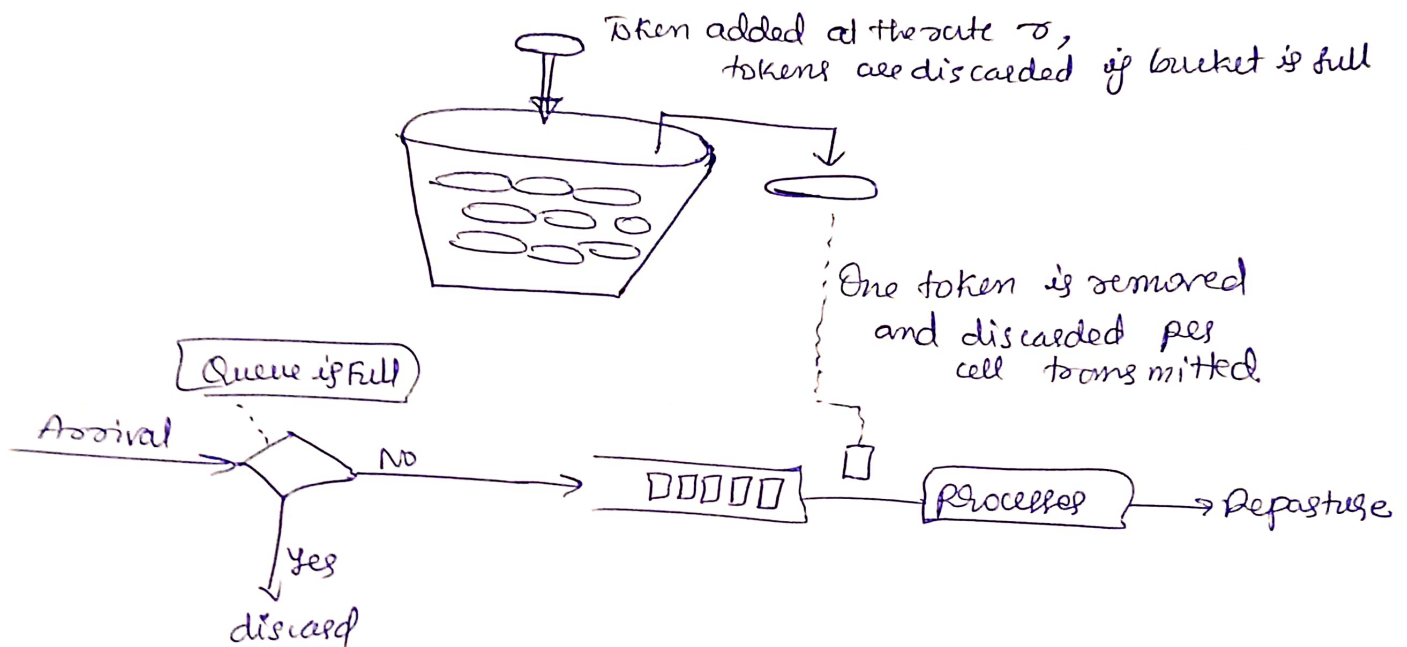
Rate of processing is remain same in entire data communication where not have any control in incoming traffic but we can have outgoing is constant.





ii) Token bucket Algorithm:

Whenever a particular network is empty, it gains some amount of coins and it will be filled to token buckets.



Whenever a node has tokens available in bucket it can start data communication. Whenever data is communicated the token will be removed.

→ Integrated Services (INTSERV)

Traditional internet provided common services to all applications.

To provide different quality of service for different applications running on internet, proposed model as Integrated services.

- Integrated service model is flow based model.

where for every pair of sender and receiver the resources are reserved throughout the transmission media.

Resources are → Memory, Bandwidth.

3 steps :-

i] data : Identify the quality of service to depending on data.

ii] Scheduling : By various scheduling algorithms are used to forward the packets.

iii] Admission control : Permission need to every router to provide necessary quality of service.

If not able to provide then data transmission stops.

Services :-

↳ Guaranteed service : whenever not able to accept any amount of delay.

↳ Controlled load service : some amount of delay accepted where communication is not that so important.

→ Resource reservation protocol :-

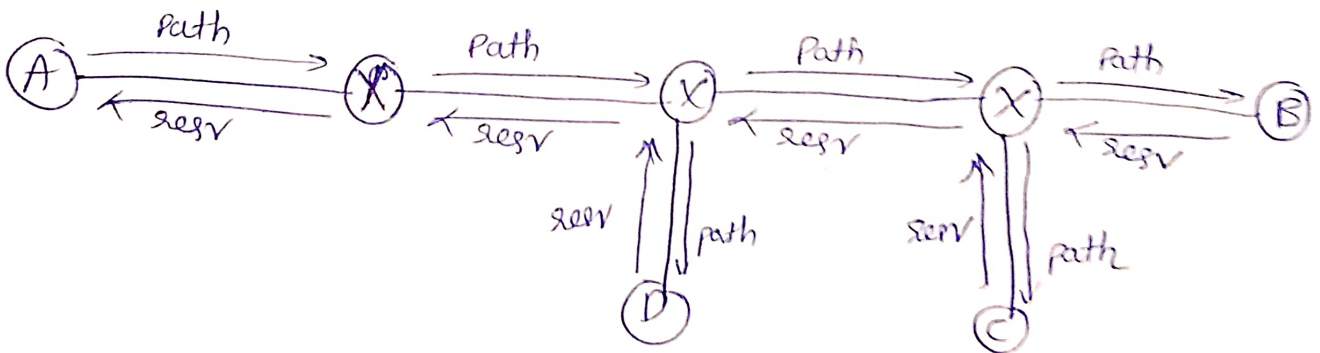
- To reserve resources we need to know about the path to transmit the data and reserve resource.
- IP is a connectionless, but RRP is built on top of IP to make it connection oriented and to reserve resources.
- RRP is mainly implemented for multicasting and unicasting they consider as special case of multicasting.
- Receivers will be reserving the resources.

RSVP messages :-

There are n messages in RRP but only 2 messages are considered and elaborated.

- Path
- resv

Consider the network scenario, where network layer identifies best path sender will send the path message to destination to transmit the data,



i] Path :-

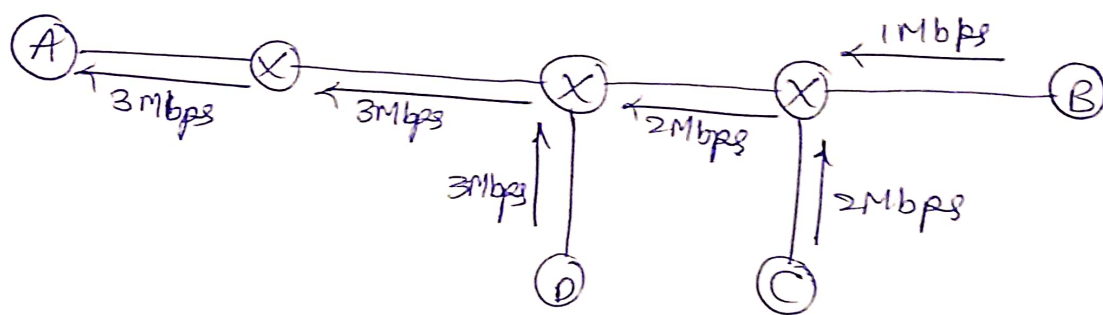
path message will contain two information

↳ best path

↳ resources needed for transmission

ii] The destination will send the Resv message to sender through the same bestpath and to store/reserve all the resources needed in router.

- consider from B bandwidth needed is 1Mbps but from C it need 2Mbps and D is 3Mbps, in this case the maximum among will be reserved as resource.



→ In RRP, soft state is about storing resources need to reserved at every pair of sender and receiver, this information is maintained at router and this is dynamic so it is called soft state.

- If router not updated dynamically then it is hardstate.

⇒ Drawbacks of Integrated service models :-

i] Scalability : Internet contains billion of devices, not able to provide memory to router to store huge amount of information.

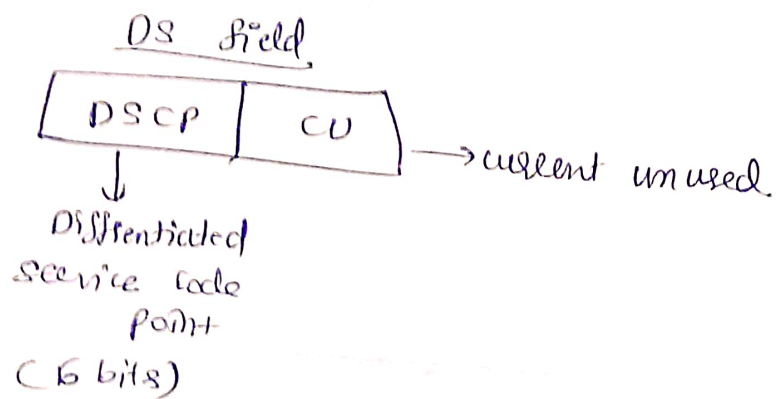
ii] Limited service class : It provides only 2 service class

Differentiated Service model:-

To overcome the drawbacks of INTSERV (Integrated service models),

- i] In differentiated service model the overhead of identifying the QoS is transferred from model to sender.
- ii] The memory ~~not~~ needed at every router is very less.

To provide information of resource reservation in header one more field added as DS field it stands for differentiated service.



Peshop behaviour:-

DSCP include peshop behaviour information.

How the router needs to behave when it is receive packet from sender.

- i] Default behaviour : Every common application not worry about delay & bandwidth.
- ii] Augmented behaviour : Ensure to provide less delay and less packet loss and good bandwidth.
- iii] Assured → When it give guarantee that no packet loss.

Chapter - 2

Applications of application Layer:

i] World wide Web :

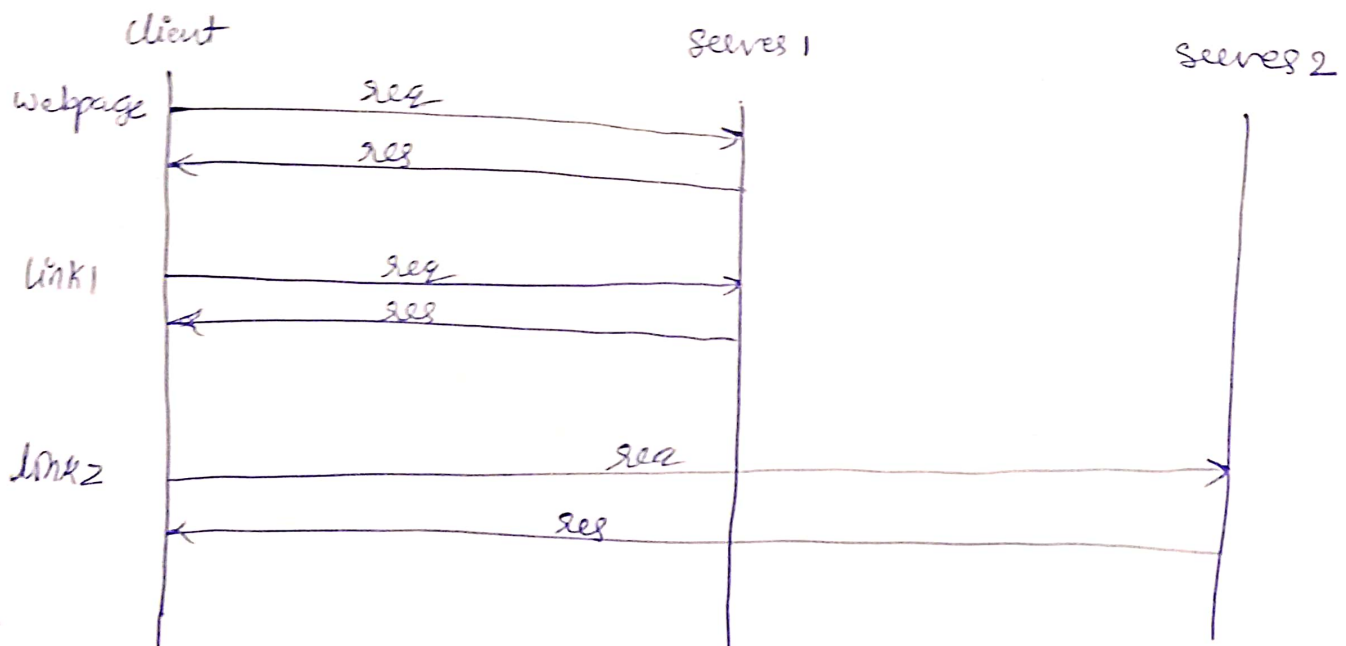
In older days it was used only for research purpose.

Web is a collection of information, information present in the form of webpages and distributed and related webpages are linked together.

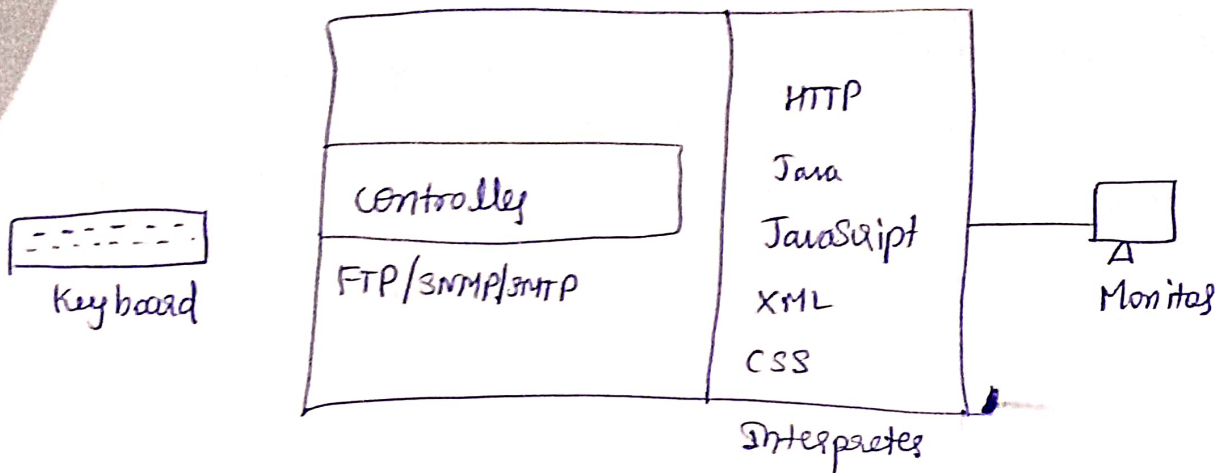
These are collection of documents will be stored in a distributed server where this feature is distributed.

The every webpage will have ~~ph~~hyperlink to navigate the other webpage so it is called linked, and transition happens for audio, video other files so it is also called as hypermedia.

Web uses a client server architecture



Web client (browser) :-



server sends the responses and it will be in cache memory because it may again request the same res in future, for some amount of time it will be kept in cache memory.

URL (Uniform Resource Locator):

- ↳ protocol
- ↳ domain name/ip address
- ↳ port no
- ↳ path

Syntax: protocol : // domainname : port / path

http : // www.google.com / sit

Web documents / webpages :-

- i] Static : Fixed content documents that are created and stored in a server. If client send n request all n responses will be same website.

These prepared by HTML, XML, XSL, XHTML

i) Dynamic : These documents are created by a web server whenever a browser requests the document.

When a request arrives, the web server runs an application program or script that creates a dynamic document.

Scripting languages : Java server pages (JSP), ASP.

iii) Active documents

For many applications we need a program or a script to be run at the client site. These are called active documents.

When a browser requests an active document, the server sends a copy of the document or a script. The document is then run at the client site.

One way to create an active document is to use Java applets.

HTTP [Hypertext transfer protocol]:

It is used to access webpages from servers. It will follow client server architecture.

It makes use of TCP protocol, it is connection oriented and reliable protocol.

The client and server, however, do not need to worry about errors in message exchange and loss of any message, because TCP is reliable and will take care of this matter.

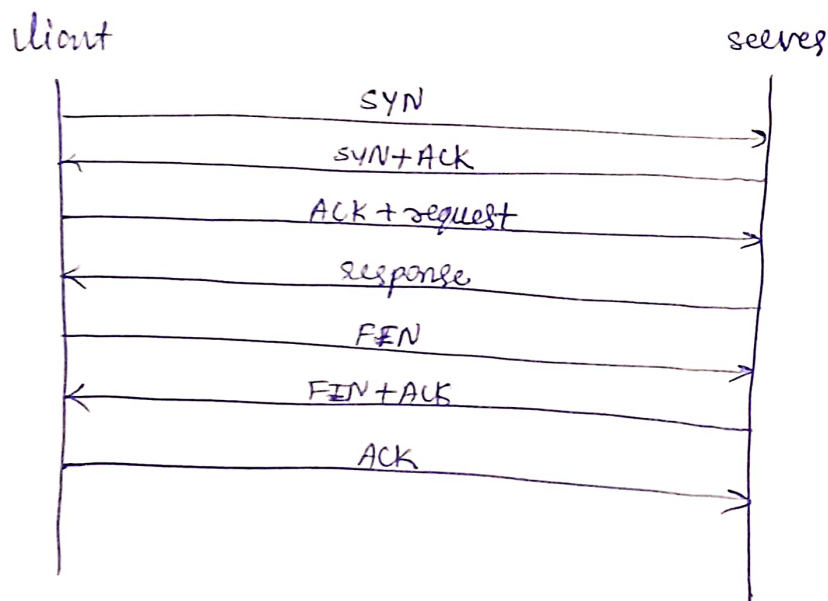
i) Non persistent connection:-

In a non persistent connection, one TCP connection is made for each request/response.

The following steps in this strategy:-

- 1] The client opens a TCP connection and sends a request
- 2] The server sends the response and closes the connection
- 3] The client reads the data until it encounters an end of file marker; it then closes the connection.

It follows single request response.

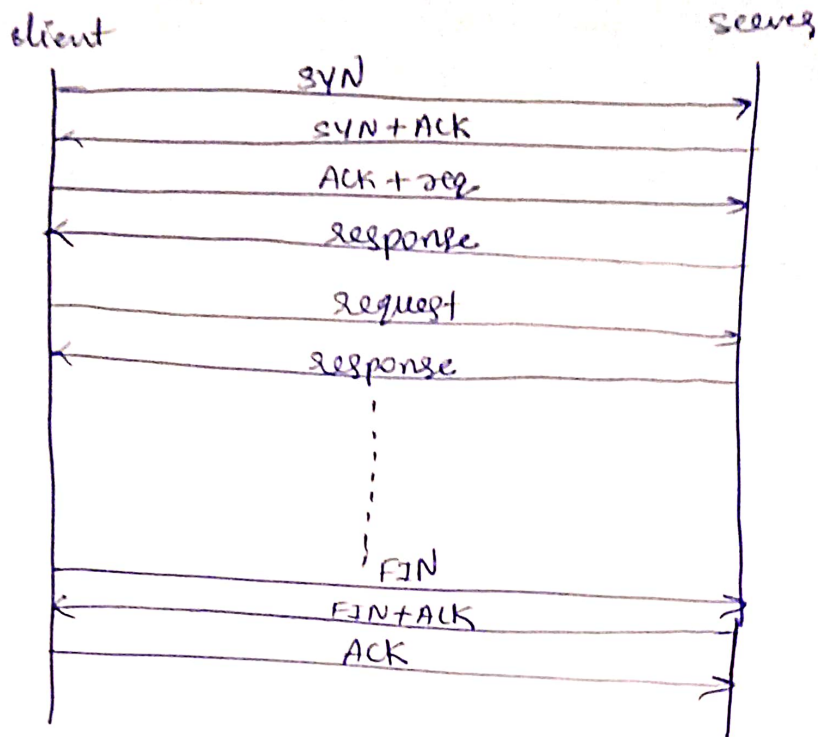


ii) Persistent connection:-

HTTP version 1.1 specifies a persistent connection by default.

In a persistent connection, the server leaves the connection open for more requests after sending a response.

The server can close the connection at the request of a client as if a time out has been reached.



HTTP Message formats:-

Request:-

Request line

Method	sp	URL	sp	version
--------	----	-----	----	---------

Header line

Header name	sp	value
-------------	----	-------

Blank line

--

Body

variable number of lines

method :

- GET → webpage
- HEAD → Information about webpage
- POST → client is sending webpage to server
- TRACE → debugging

SP → space

URL → Uniform resource locator

version → 1.1

Header line → Includes additional information

Blank line → is contain no information, used to separate body and header.

Response :-

Status line

version	sp	status code	sp	phrase
---------	----	-------------	----	--------

Header line

Header name	sp	value
-------------	----	-------

⋮

Blank line

--

Body

variable number of lines

status line : 100 → information regarding the response

200 → Successful request and response between client & server.

300 → It is indicating that current server not able to give webpage, it is forwarding to some other server by page.

400 → Error occurred at client side

500 → Error at server side

phrase : Information regarding the status code mention in response.

conditional request: If a client wants webpage by specifying the condition in a header line.

Cookies :- stateless : Information not maintained b/w client & server

statefull : To maintain information the cookies are created and they are shared b/w client & server

cookies are created by server

Information present inside the cookies :

- Websites are being used as electronic storage
- websites need to allow access to registered clients only
- websites used as portals
- Websites are just advertising agencies.

Proxy server :-

A proxy server is a computer that keeps the copies of responses to recent requests. The HTTP client sends a request to the proxy server. The proxy server checks its cache, if response not stored then it will send that request to server.

The proxy server reduces the load on the original server, decreases traffic and improves latency.

Security of HTTP protocol :-

HTTP will not provide any security but it will run on HTTP secure socket layer (SSL).

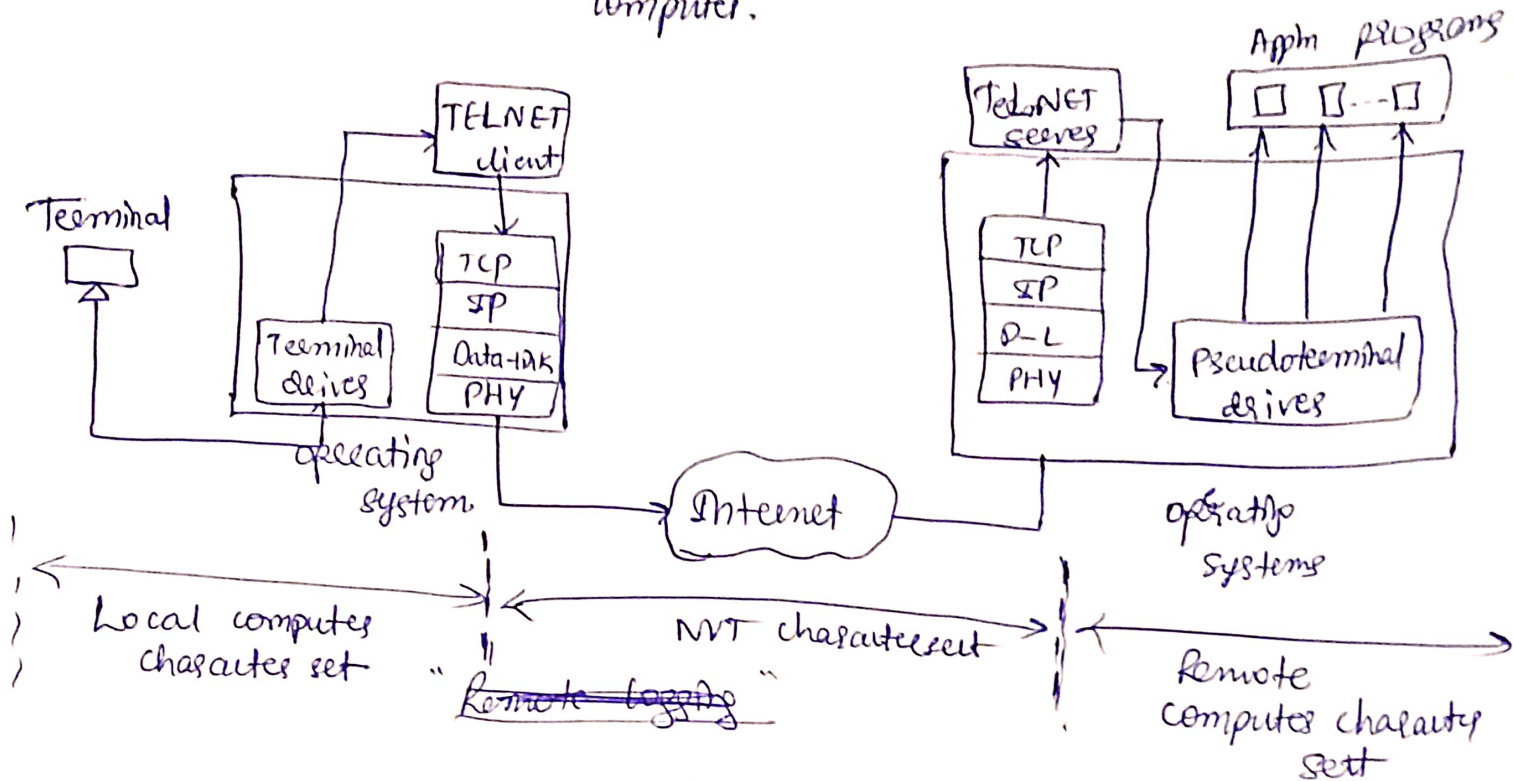
HTTPS is secured version of HTTP

↳ Hyper text transfer protocol secure.

TELNET [Terminal Network] :-

It is mainly used for remote login.

Remote login : To access ~~page~~ program running on different computer.



NVT → Network virtual terminal : It will act as translator b/w two network devices

NVT act as translator.

TELNET not provides not security because TELNET while login to remote system, in a transmission media anybody can access username & password.

SSH (Secure shell) is used securely across the remote login.

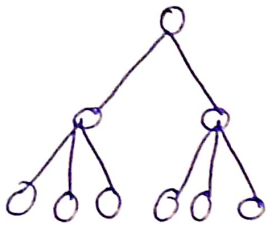
DNS [Domain Name System]

- As human being it is not easy to remember multiple ip at a time, because of this along with ip, domain name will also be there.
- Domain names are used only for internet and doesn't know about IP, where DNS will map domain name to IP

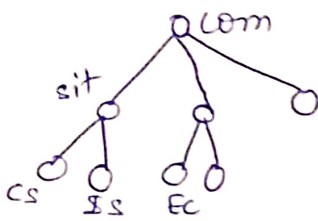
Namespace : It contains mapping b/w ip address and domain name

IP address $\longleftrightarrow$ domain Name

Domain Name space :- It will be maintained in a tree structure.



A tree maximum of 128 level in a single server. Every node in level have label, label is a string, where it contains maximum of 63 characters.



cs.sit.com

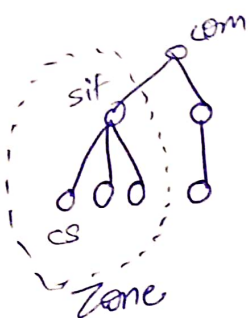
(Fully qualified domain name)

Flat domain name $\rightarrow$ Not maintaining the structure

(Hierarchical structure)

There is a possibility of duplicates.

cs.sit $\rightarrow$ partially qualified domain name
where child not reach the root.

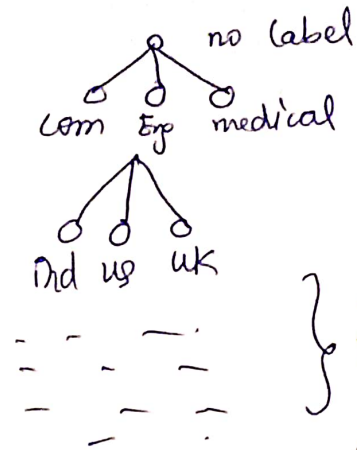


Zone $\rightarrow$ the part of tree structure particular node is responsible

DNS in Internet:-

Tree structured will be maintained with respect to two things

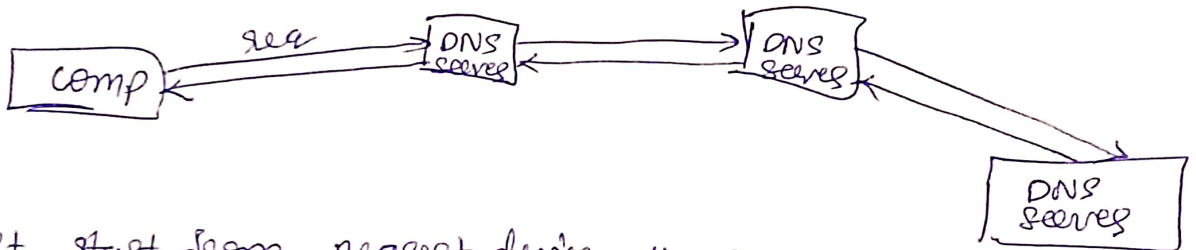
- i) global domain
- ii) country domain



} remaining part left to who is creating the tree structure.

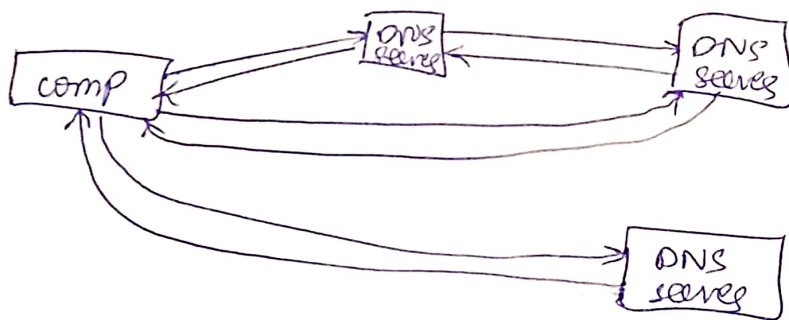
Resolution: Finding the domain name for a particular IP address.

i) Recursive resolution:



A request start from nearest device the response will forwarded to client untill it is recursively forwarded to n DNS servers,

ii) Iterative resolution:-



Format in which DNS send domain name

resource records :-

Domain name	type	class	TTL	values
				↓ name

Domain name - specifies which domain name

type - fully / partially qualified

class - country

TTL - number of intermediate devices

values - corresponding domain

Registrar - It is an organisation to contact to add a new node, which is created by IETF

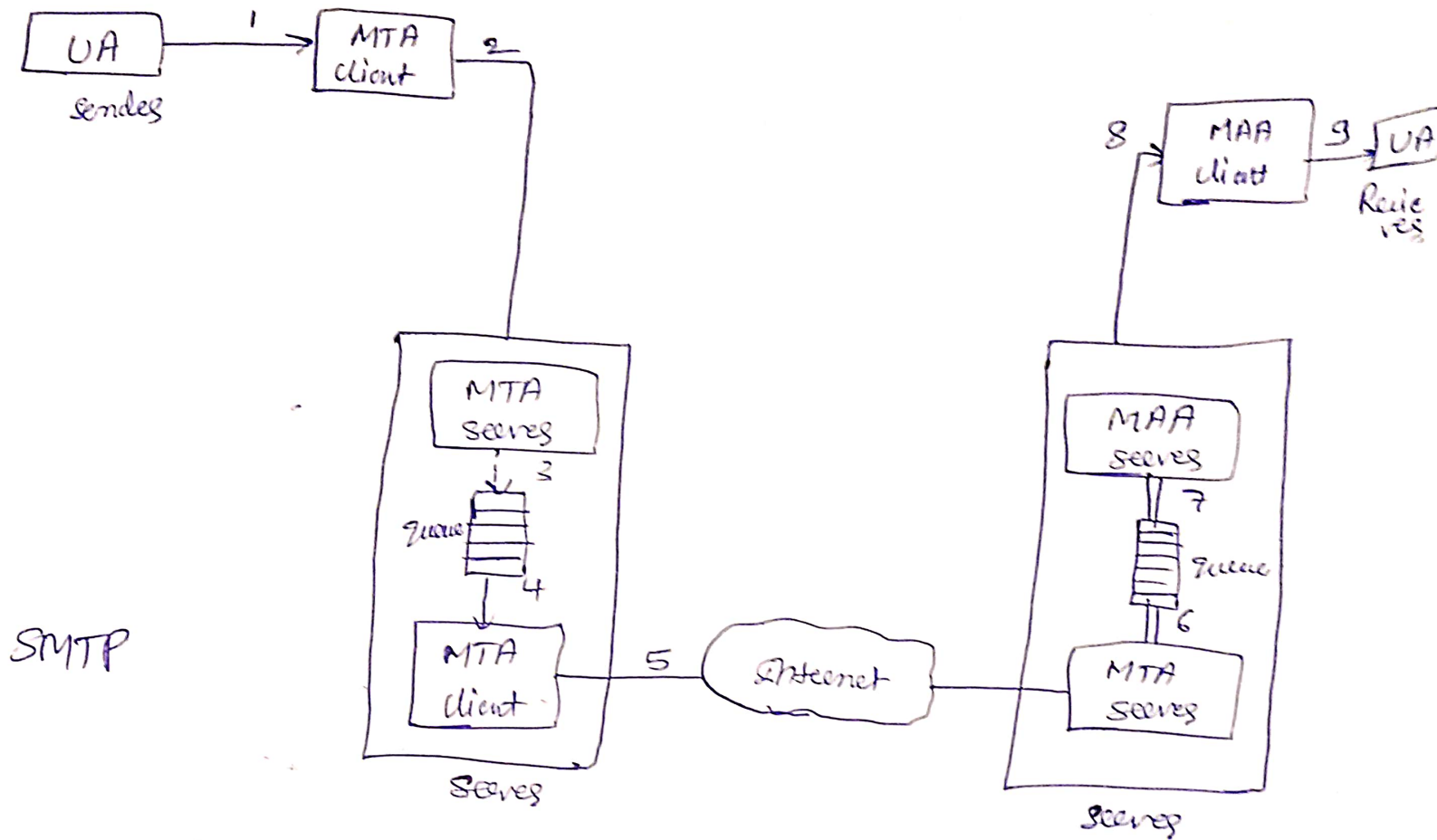
Dynamic DNS (updates) To update in domain name we will contact dynamic DNS, which is DDNS.

DNS security protocol :-

- Traditional DNS will not secure, It provides new protocol DNS security protocol (DNSSEC).

Electronic Mail :-

Architecture :- Whenever send a mail but not worry about the receiver and send a message.



UA → user agent, it present at both sender and receiver, it used compose the client

MTA → Message transfer agent

SMTP → Simple mail transfer protocol

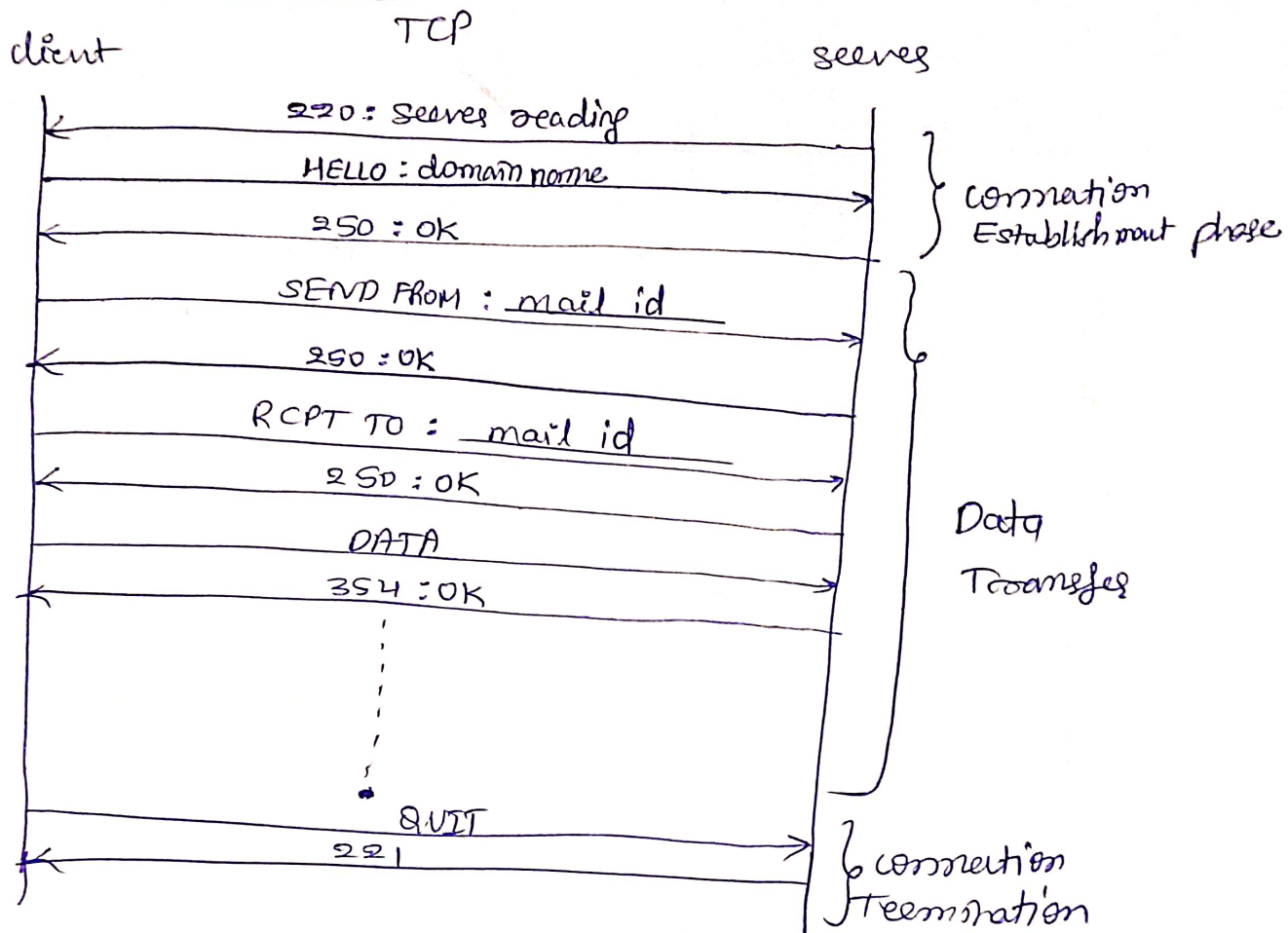
MAA → Message access agent

SMTP to have communication

Client
↓
Keyword : args

Server
Response
3 digit code : text

Example communication:



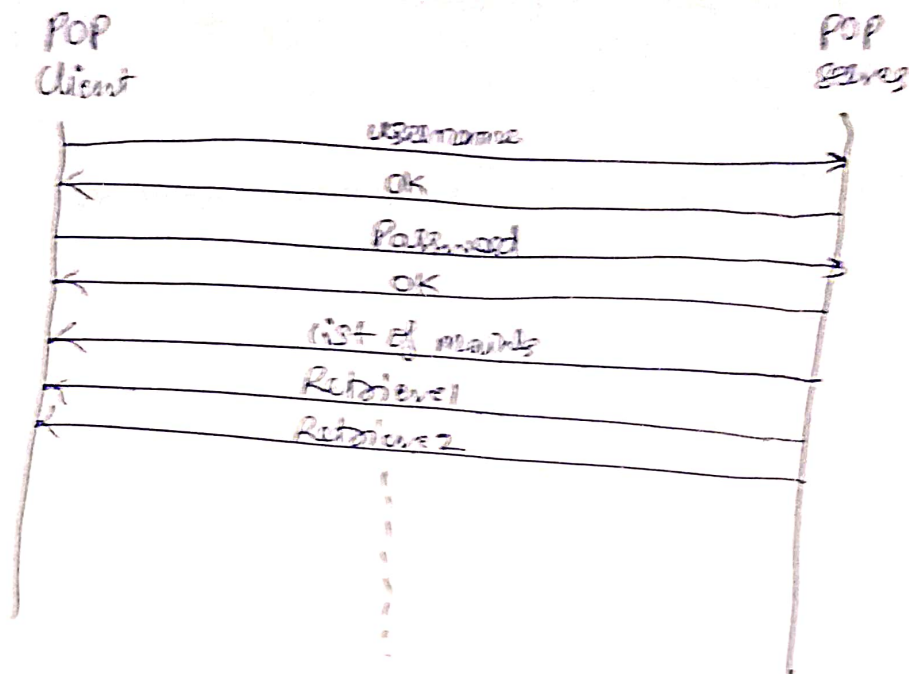
- Sending 220 code where server is ready and the when it is not ready it will show 421

- SMTP uses TCP for its communication.

MUA (to implement we have 2 different protocols)

i) POP3 → Post office protocol version 3

ii) IMAP4 → Internet mail access protocol version 4



IMAP:

- i] Search
- ii] Partially download content
- iii] Check for subject of mail
- iv] Possible to rename mailbox, delete and create.

Web based mail : In web based mail instead of MTA we can replace with HTTP at client & server.